

Dataset

The UCR Earthquake dataset is a binary time series classification benchmark. Each sample is a seismic waveform signal, and the task is to classify whether the signal corresponds to an earthquake event or background noise.

The dataset exhibited a significant class imbalance:

- Class 0 (background noise): 264 samples
- Class 1 (earthquake): 58 samples

Data Augmentation

Due to the class imbalance, a targeted augmentation strategy was applied to the training set only. The test set was left untouched to ensure unbiased evaluation. The minority class was augmented aggressively using strong augmentation, while the majority class received light augmentation to add diversity without over-representing it.

For each minority-class sample, 6 strongly-augmented copies were generated. Majority-class samples each received 1 lightly-augmented copy.

1 Jitter

Adds Gaussian noise independently to each time step. Simulates sensor noise and real-world measurement variability, helping the model generalise to noisy inputs.

$$x'(t) = x(t) + N(0, \sigma^2) \quad \sigma = 0.03$$

2 Scaling

Multiplies the entire signal by a random scalar drawn from a normal distribution. Accounts for amplitude variations across different sensors or recording conditions.

$$x'(t) = \alpha \cdot x(t) \quad \alpha \sim N(1.0, 0.15^2)$$

3 Time Shift

Shifts the signal forward or backward along the time axis by a random integer offset, padding the emptied region with zeros. Teaches the model that an event may begin at different positions within the recording window.

$$x'(t) = x(t - \delta) \quad \delta \sim \text{Uniform}[-15, +15]$$

4 Time Warp

Applies a smooth, non-linear distortion to the time axis by interpolating between randomly drawn warp factors at a fixed number of knot points. This mimics natural variability in event timing and propagation speed, and is more realistic than a simple shift.

$$x'(t) = x(\varphi(t)) \quad \varphi = \text{cumsum}(\text{interp}(w_1 \dots w_k)) \quad w_i \sim N(1.0, 0.25^2)$$

Time Series to Image Conversion

To leverage image-based deep learning architectures, the raw time series signals were encoded into 2D image representations. Each method captures different structural properties of the signal. All images were generated at 128×128 pixels.

1 Gramian Angular Summation Field (GASF)

The time series is first re-scaled to $[-1, 1]$ and each value encoded as an angle on a unit circle:

$$\varphi_i = \arccos(x_i) \quad x_i \in [-1, 1]$$

The GASF matrix is then formed by computing the cosine of the sum of every pair of angles:

$$\text{GASF}[i, j] = \cos(\varphi_i + \varphi_j)$$

Because $\cos(\alpha + \beta) = \cos\alpha \cdot \cos\beta - \sin\alpha \cdot \sin\beta$, the diagonal of the matrix recovers the original time series values. GASF preserves the absolute temporal structure and is well-suited to capturing the overall trend and periodicity of the signal.

2 Gramian Angular Difference Field (GADF)

Identical encoding to GASF, but uses the sine of the difference of angle pairs instead:

$$\text{GADF}[i, j] = \sin(\varphi_i - \varphi_j)$$

$\sin(\alpha - \beta) = \sin\alpha \cdot \cos\beta - \cos\alpha \cdot \sin\beta$. The diagonal of GADF is zero by definition, and off-diagonal values encode the relative change between time steps. GADF therefore highlights the rate of change and directional transitions rather than absolute values.

3 Recurrence Plot (RP)

A Recurrence Plot encodes the pairwise similarity of the signal with itself across time. Each pixel (i, j) indicates whether the signal state at time i is close to the state at time j :

$$\text{RP}[i, j] = \Theta(\varepsilon - \|x(i) - x(j)\|)$$

where Θ is the Heaviside step function and ε is a distance threshold. Pixels are 1 (recurrent) or 0 (non-recurrent). The resulting image reveals periodic patterns as diagonal lines, drifts as fade-outs, and abrupt changes as broken structures. These features are extremely difficult to detect from the raw waveform directly.

Baseline Classification (Raw Time Series)

Before training any image-based models, classical machine learning methods were applied directly to the raw time series. This establishes a performance baseline to evaluate whether image representations offer a meaningful advantage.

- **SVM (RBF kernel)** — Finds the maximum-margin hyperplane in a kernel-transformed feature space. The RBF kernel implicitly maps input vectors to an infinite-dimensional space, allowing non-linear boundaries.
- **Random Forest** — An ensemble of decision trees trained on random feature/sample subsets. Robust to noise and provides feature importances.

- **Euclidean kNN (k=3)** — Classifies by majority vote among k=3 nearest neighbours using standard L_2 distance. Fast but sensitive to temporal misalignment.
- **DTW-kNN (k=7)** — Same as kNN but uses Dynamic Time Warping distance, which allows elastic alignment of sequences and is the de facto standard metric for time series classification.

DTW distance between two series Q and C of length n is defined recursively:

$$DTW(i, j) = |q_i - c_j| + \min \{ DTW(i-1, j), DTW(i, j-1), DTW(i-1, j-1) \}$$

Results

The table below summarizes train and test accuracy for each baseline method:

Method	Train Acc.	Test Acc.	Observation
DTW-kNN (k=7)	92.40%	74.10	Best test accuracy
Random Forest	100.00%	72.66%	Over-fit, decent test
SVM (RBF kernel)	99.46%	58.99%	Moderate generalization
Euclidean kNN (k=3)	100.00%	25.90%	Severe over-fit

Key observations from the baseline results:

- **DTW-kNN achieved the best test accuracy (74.10%)** with the lowest train accuracy among the strong baselines (92.40%), suggesting better generalisation through elastic alignment.
- **Euclidean kNN severely overfit** — perfect train accuracy but only 25.90% test accuracy, highlighting the sensitivity of L_2 distance to temporal misalignment in seismic signals.
- **Random Forest and SVM both overfit** to varying degrees. Random Forest generalised better than SVM, likely due to its ensemble structure.

CNN-Based Image Classification

A Convolutional Neural Network (CNN) was trained separately on each image representation type (GASF, GADF, RP). CNNs are well-suited for image classification because convolutional layers automatically learn spatial hierarchies — edges, textures, and higher-level patterns — directly from pixel values, without hand-crafted feature engineering.

By training one model per image type, the effect of different encoding strategies on classification accuracy can be compared in a controlled way. The hypothesis is that

image encodings capturing complementary signal structure may yield different accuracy profiles and that combining them could provide further gains.

7. Pipeline Summary

The end-to-end pipeline is as follows:

1. Load UCR Earthquake dataset and inspect class distribution.
2. Apply augmentation (jitter, scaling, time shift, time warp) to the training split only to balance classes.
3. Convert training and test samples into GASF, GADF, and RP images at 128×128 pixels.
4. Train classical ML classifiers (SVM, Random Forest, Euclidean kNN, DTW-kNN) on raw time series as a baseline.
5. Train CNN models on each image type and compare results against the baseline.